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Worksheet – Heat transfer simulation

Grade 6 · Physics · Unit 1 Thinking about Energy · Heat and temperature, conduction, convection, radiation

Name: _____ Class: _____ Date: _____

Open the simulation (scan the QR code), do the 4 stations with Marie and fill in this sheet as you go. Write temperatures with the unit °C.

school.taneski.com/science/g06/06-heat-transfer/

Station 1 · Heat or temperature? (1A Heating race · 1B Big and small)

1 Heating race: both beakers get the same heat for 5 minutes. Temperature after 5 minutes: small beaker (100 g) _____ big beaker (400 g) _____

2 Why did the big beaker warm up less? Use the word **particles**.

3 Big and small: both at 60 °C, the same ice cube in each. After 5 minutes: small cup _____ big beaker _____. Where did more ice melt? _____

4 A match flame is about 600 °C and a bath is 40 °C. Which has more thermal energy? _____ Why?

5 Temperature is measured in _____ with a _____. Thermal energy (heat) is measured in _____. Heat always moves from _____ to _____.

Station 2 · Conduction – the metal star

6 Copy the times from the results table.

Metal	Copper	Aluminium	Brass	Nickel	Steel
Time for the pin to drop (s)					

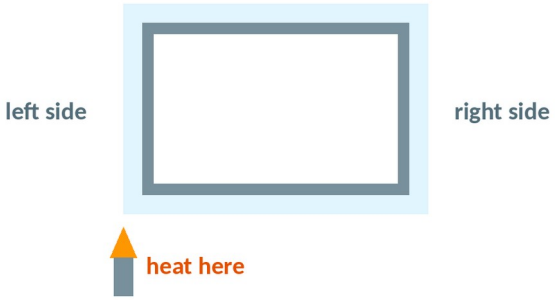
7 The best conductor is _____ and the worst of the five is _____, because _____.

8 Draw 6 particles in each box (look at the particle model). Show how much they vibrate with little lines.

Cool end of the arm	Hot centre of the star

9 Conduction: the particles _____ more and bump into their _____. The particles stay in their places – only the _____ moves along.

Station 3 · Convection – the convection tube



10 On the picture, draw arrows to show how the purple colour moved.

11 Why does the hot water rise? Use: **particles · spread out · less dense**

_____

_____

_____

12 Convection happens in liquids and _____, but not in _____, because _____.

Station 4 · Radiation (4A Three bottles · 4B Leslie cube)

13 Bottles under the infrared lamp – temperature of the air after 6 minutes:

Bottle	Black	White	Shiny silver
Temperature after 6 min (°C)			

14 The best absorber is the _____ bottle. The silver bottle stayed coolest because _____.

15 Leslie cube full of 80 °C water – infrared thermometer readings:

Face	Matt black	Matt white	Dull silver	Shiny silver
Reading (°C)				

16 Sam says: “The black face is hotter than the shiny face.” Is Sam right? Explain.

_____

_____

_____

Summary · three ways heat moves

	Conduction	Convection	Radiation
Does it need particles? (yes / no)			
Where does it happen? (solid, liquid, gas, vacuum)			
An example from the simulation			

17 One thing I learned today:

_____

_____